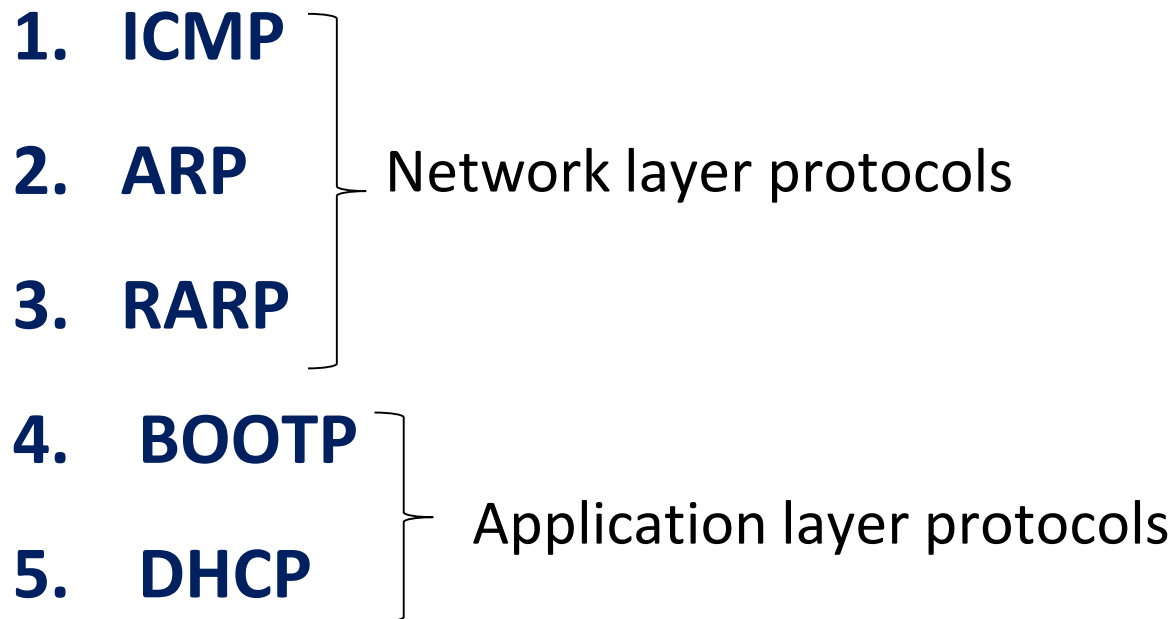


Unit II

Internet Control Protocols

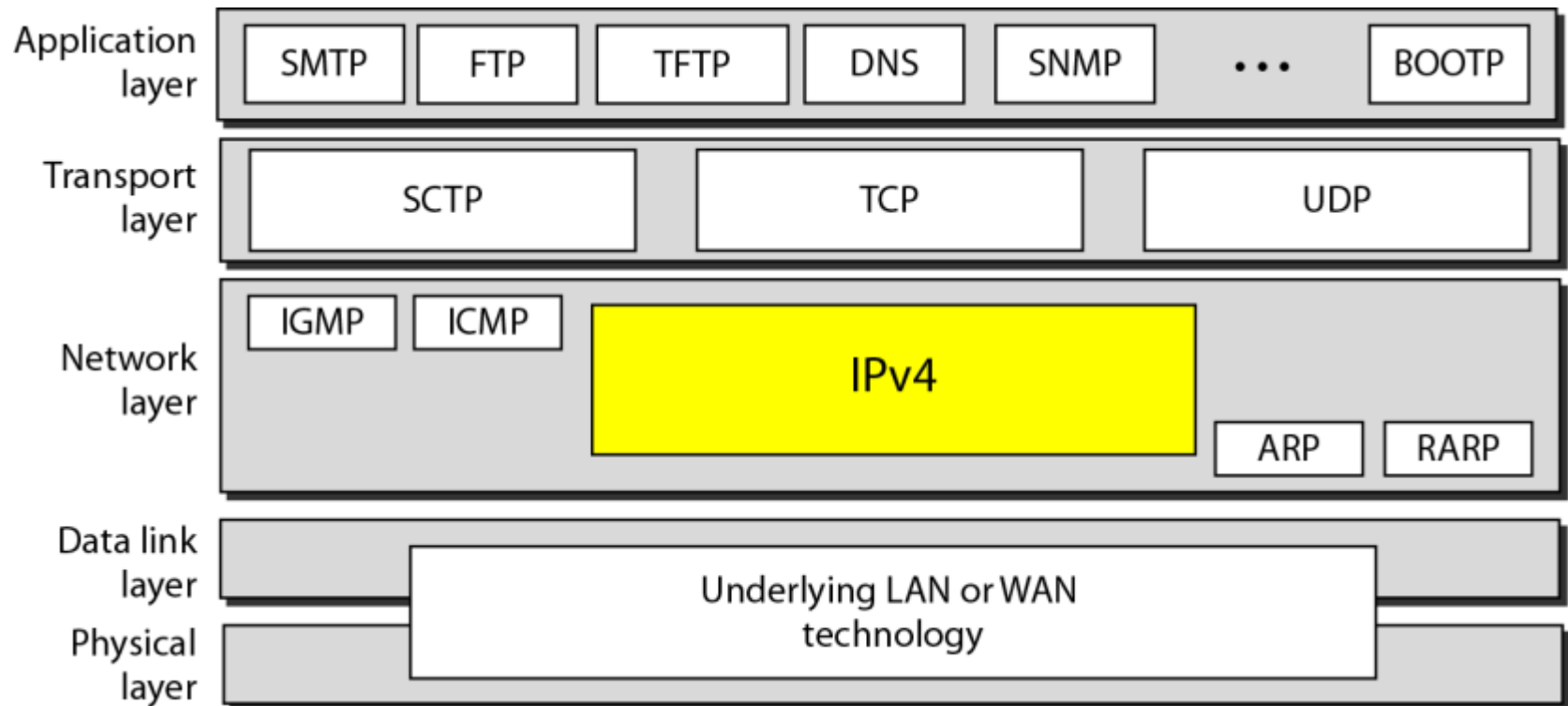
Internet Control Protocols

- The Internet has several control protocols used in the network layer



Internet Control Protocols

Position of IPv4 , ARP,RARP, ICMP and IGMP



The Internet Control Message Protocol (ICMP)

- Used primarily by routers to report delivery or routing problems to original source. Also used to test the Internet

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo request	Ask a machine if it is alive
Echo reply	Yes, I am alive
Timestamp request	Same as Echo request, but with timestamp
Timestamp reply	Same as Echo reply, but with timestamp

The principal ICMP message types

[www.iana.org/assignments/icmp-parameters.](http://www.iana.org/assignments/icmp-parameters)

ICMP

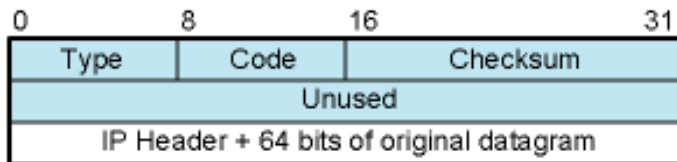
- Internet Control Message Protocol - RFC 792
 - All IP implementations should also implement ICMP
- Transfer of (control) messages from routers-to-hosts and hosts-to-hosts
- Feedback about problems
 - e.g. datagram discarded, router's buffer full
- Encapsulated in IP datagram
 - Not reliable

ICMP

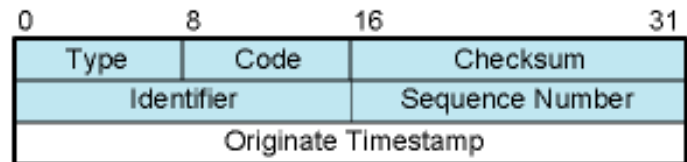
- To make efficient use of the network resources, IP was designed to provide unreliable and connectionless best-effort datagram delivery service. As a consequence, IP has no error-control mechanism and also lacks mechanism for host and management queries.
- A companion protocol known as Internet Control Message Protocol (ICMP), has been designed to compensate these two deficiencies.

- ICMP messages can be broadly divided into two broad categories: error reporting messages and query messages as follows.
- **Error reporting Messages:** Destination unreachable, Time Exceeded, Source quench, Parameter problems, Redirect
- **Query:** Echo request and reply, Timestamp request and reply, Address mask request and reply

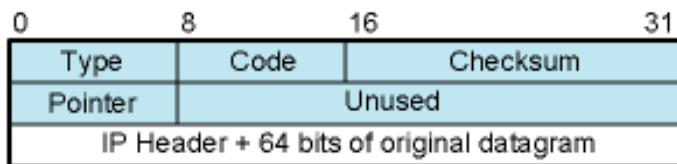
ICMP Message Formats



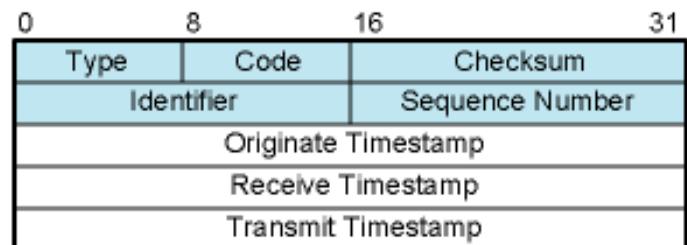
(a) Destination Unreachable; Time Exceeded; Source Quench



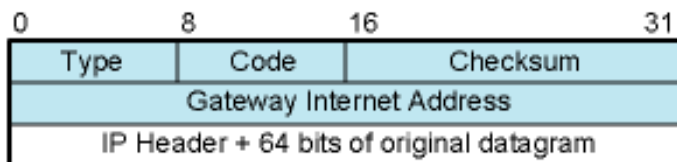
(e) Timestamp



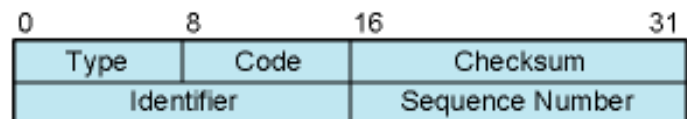
(b) Parameter Problem



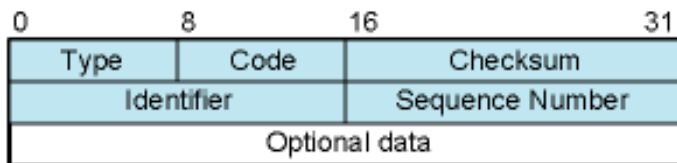
(f) Timestamp Reply



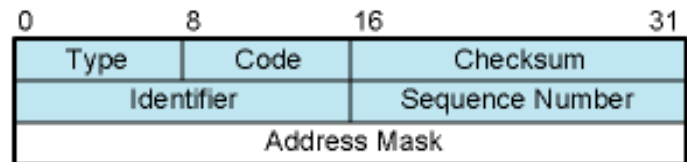
(c) Redirect



(g) Address Mask Request



(d) Echo, Echo Reply

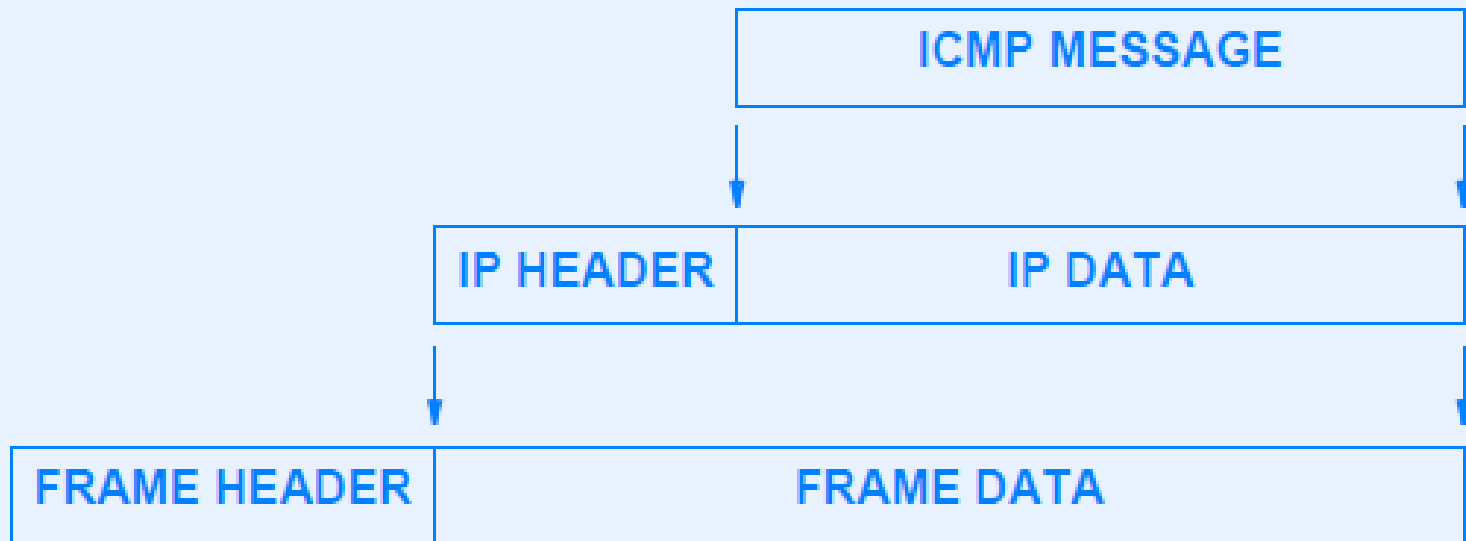


(h) Address Mask Reply

The Internet Control Message Protocol (ICMP)

- Uses IP to carry control messages

ICMP Message Encapsulation



ARP—The Address Resolution Protocol

- MAPPING IP ADDRESSES TO PHYSICAL ADDRESSES
- A standard for dynamic address resolution in the Internet
- Required , because data link layer hardware does not understand IP addresses

ARP—The Address Resolution Protocol

Example

- Computers A and B on same network
- Application on A generates packet for application on B
- Protocol software on A must use B's hardware address when sending a packet

ARP—The Address Resolution Protocol

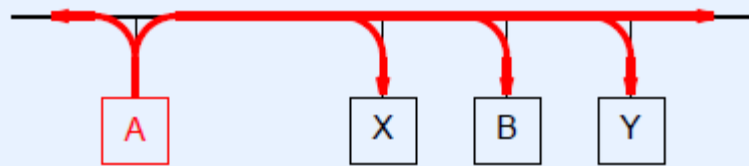
How A knows B's hardware address(MAC) ?

Simple solution is for A is to broadcast packet □ ARP packet

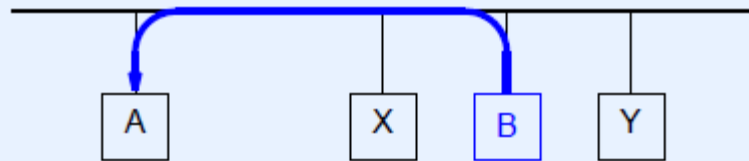
- Machine A broadcasts ARP request with B's IP address
- All machines on local net receive broadcast
- Machine B replies with its physical address
- Machine A adds B's address information to its table
- Machine A delivers packet directly to B

ARP—The Address Resolution Protocol

Illustration Of ARP Request And Reply Messages



A broadcasts request for B
(across local net only)

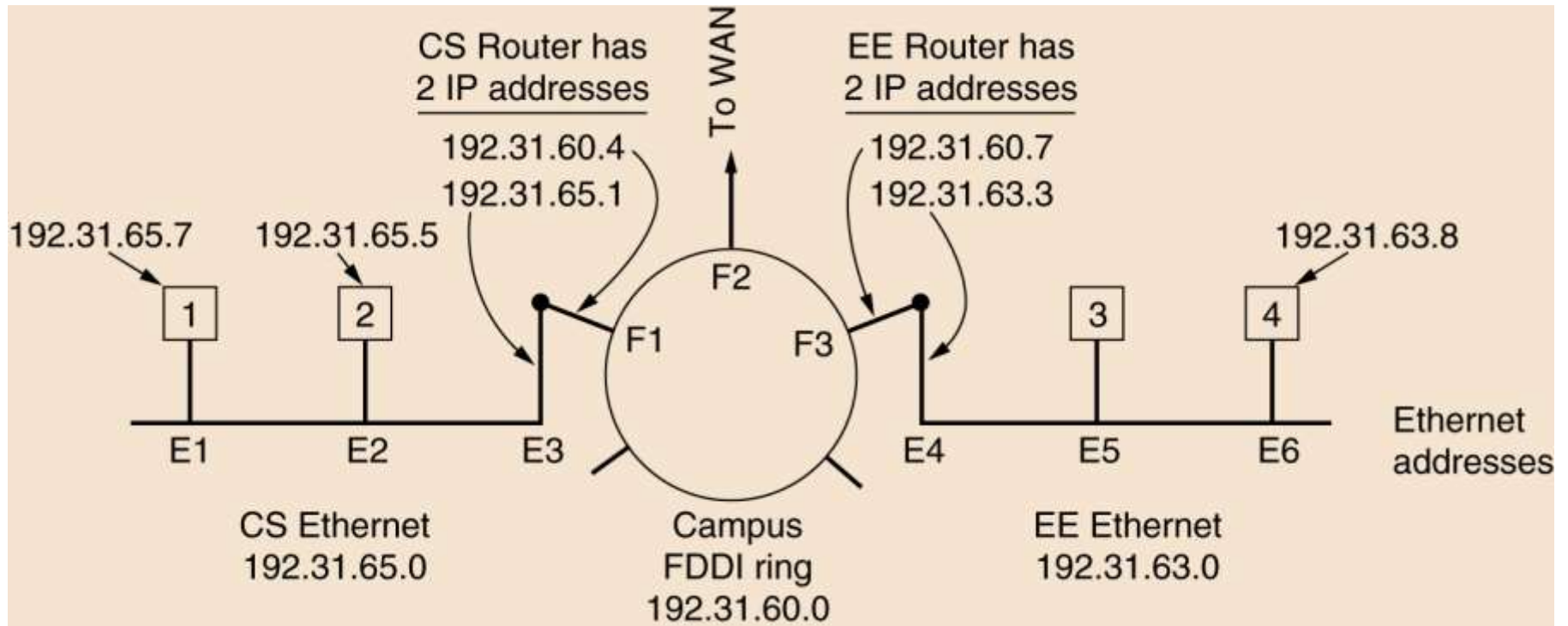


B replies to request

The protocol used for asking for physical address and getting the reply is called ARP .

ARP—The Address Resolution Protocol

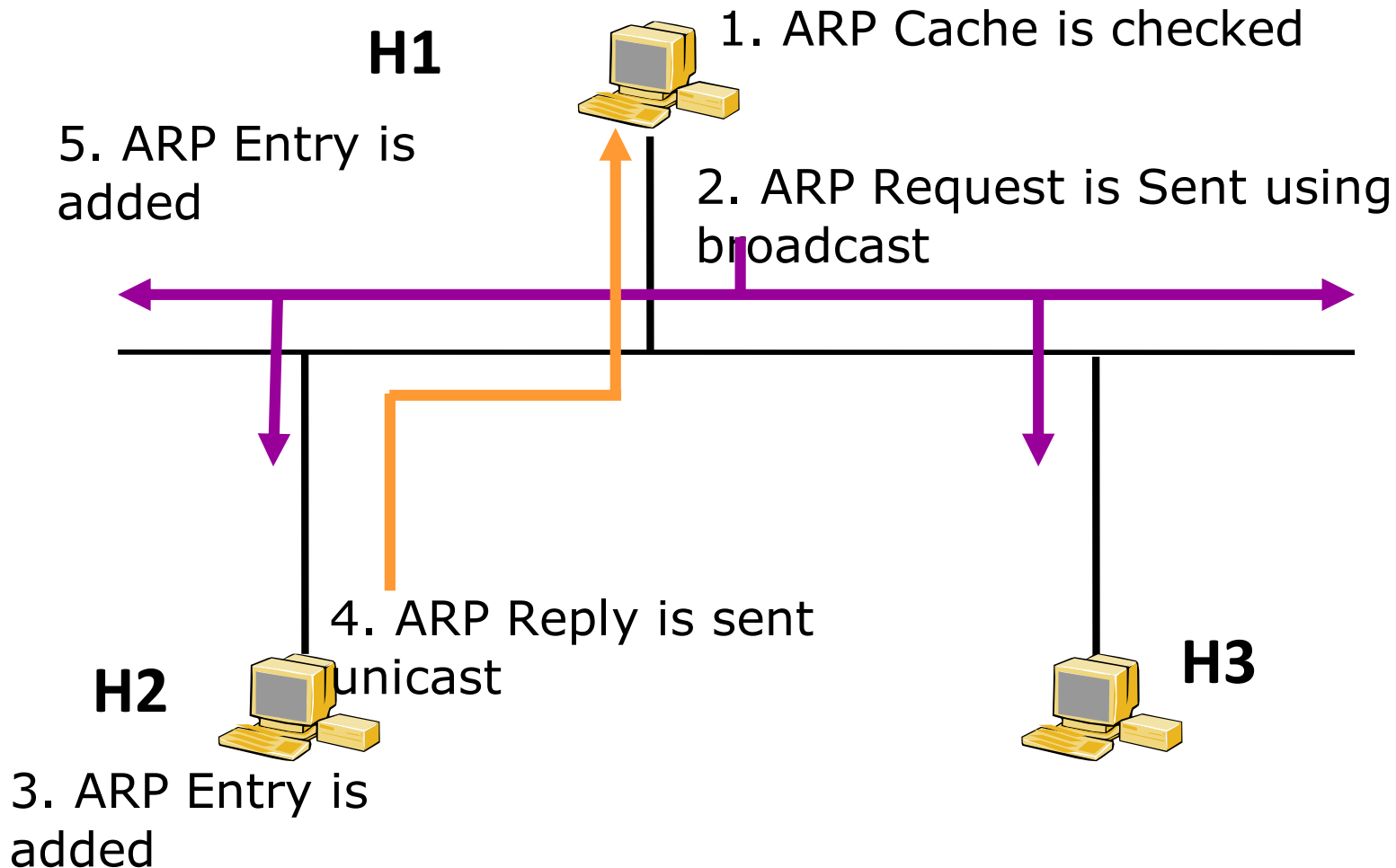
A Typical Example :



Three interconnected /24 networks: two Ethernets and an FDDI ring.

Optimizations to ARP

Caching the result to avoid a second broadcast



Optimizations to ARP

Have every machine broadcast its mapping when it boots

In the previous diagram

- When host1 (H1) broadcast, it includes its IP-to-Ethernet mapping in the ARP packet.
- When the ARP broadcast arrives at host 2 (H2), the mapping is entered into host 2's ARP cache for future use
- When the ARP reply arrives at host 1 (H1), the H2's mapping is entered into host 1's ARP cache for future use

Optimizations to ARP

Caching the result to avoid a second broadcast

In the previous diagram

- When host1 (H1) broadcast, it includes its IP-to-Ethernet mapping in the ARP packet.
- When the ARP broadcast arrives at host 2 (H2), the mapping is entered into host 2's ARP cache for future use
- When the ARP reply arrives at host 1 (H1), the H2's mapping entered into host 1's ARP cache for future use

Optimizations to ARP

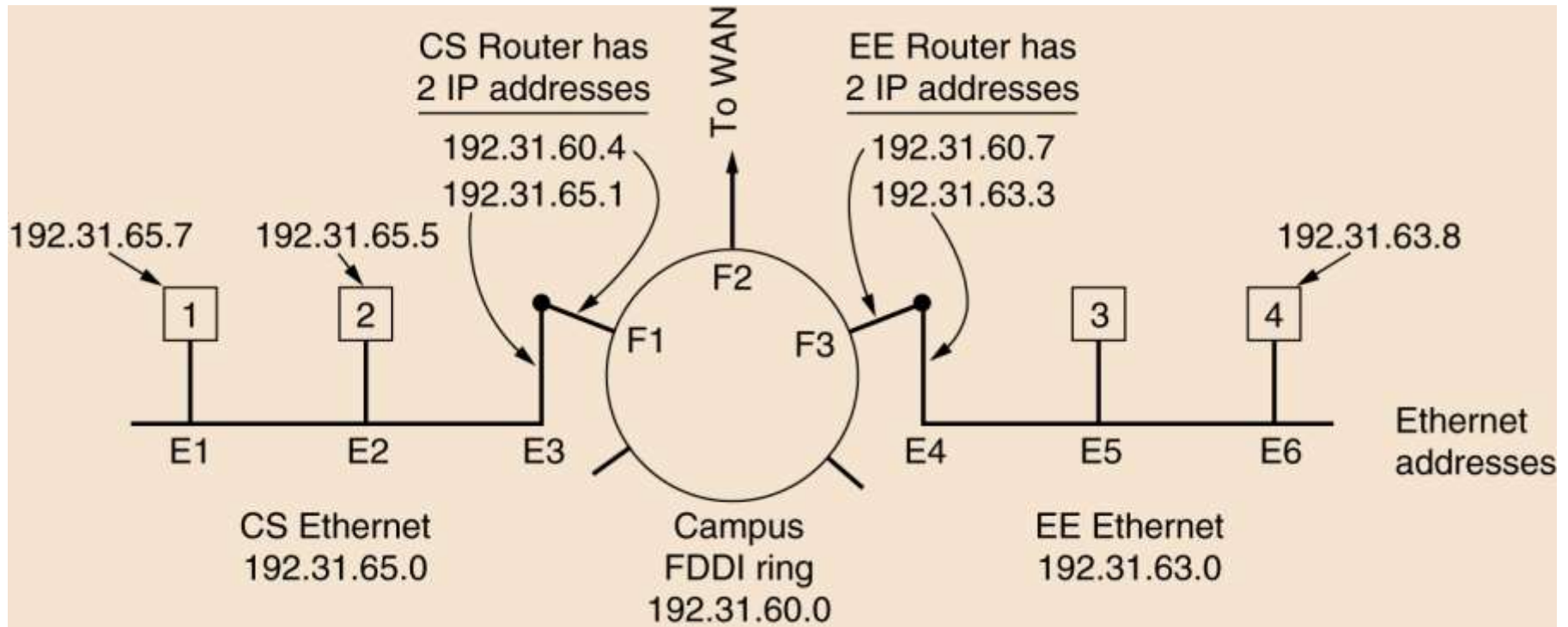
ARP Caching Table

IP Address	Hardware Address	Age (Sec)
192.168.0.2	08-00-20-08-70-54	3
192.168.0.65	05-02-20-08-88-33	120
192.168.0.34	07-01-20-08-73-22	43

Entries in the ARP cache should time out after a few minutes

Proxy ARP

If host 1 wants to send a packet to host 4 (192.31.63.8).?



ARP will fail because , it works within a LAN

Mapping Physical to IP Address

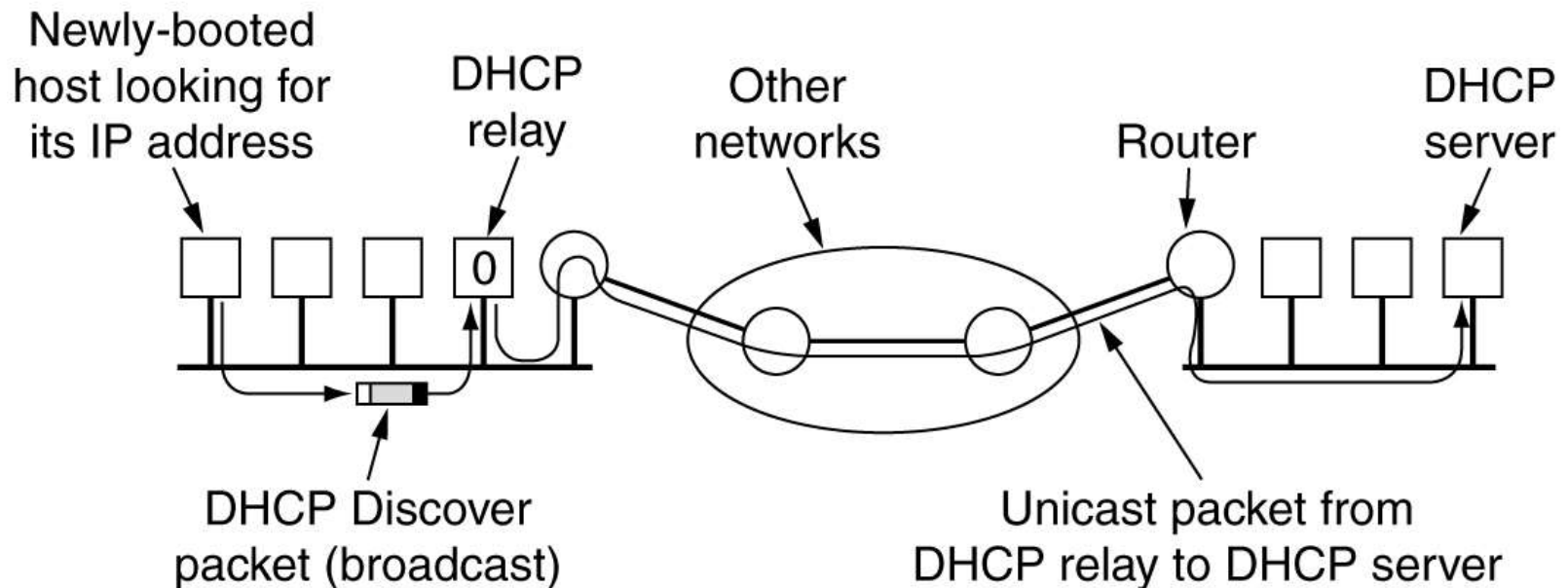
Given an Ethernet address, the corresponding IP is obtained by using following protocols

1. **RARP (Reverse Address Resolution Protocol)**
2. **BOOTP (Bootstrap Protocol)**
3. **DHCP (Dynamic Host Configuration Protocol)**

DHCP

- Allows both manual IP address assignment and automatic assignment.
- **DHCP server need not be on the same LAN as the requesting host.**

A DHCP relay agent is needed on each LAN



DHCP

- **Operation of DHCP.**

1. A newly-booted machine broadcasts a **DHCP DISCOVER** packet
2. DHCP relay agent on its LAN intercepts the DHCP DISCOVER packet, it sends the packet as a unicast packet to the DHCP server

the relay agent needs the IP address of the DHCP server

OSPF— Open Shortest Path First

The Interior Gateway Routing Protocol

- *Became a standard in 1990*
- *Most router vendors now support it*

OSPF

- One of the most widely used link-state routing protocols.
- “Open”
 - *nonproprietary standard*
 - *created under the auspices of the IETF*
- “SPF”
 - *an alternative name for link state routing*

Challenges for OSPF

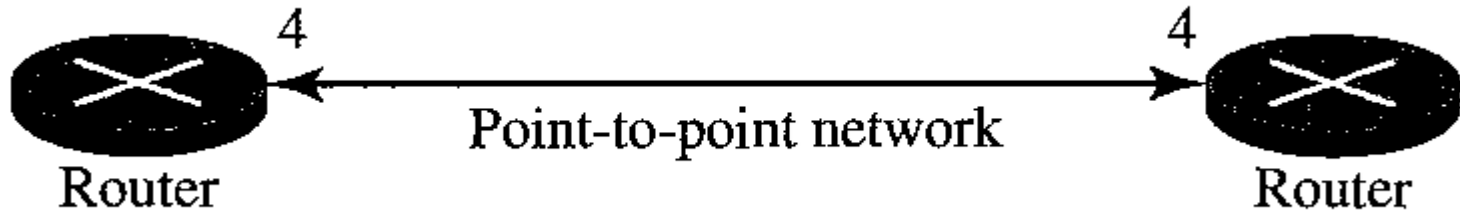
1. Supporting a variety of distance metrics.
2. Had to be a dynamic algorithm
3. Supporting routing based on **type of service**
4. **Load balancing:** splitting the load over multiple lines
5. Supporting hierarchical systems :
 - to make systems more scalable
6. security to prevent dedicated malicious users
7. Supporting tunneling.

OSPF

- **OSPF supports three kinds of connections and networks**
 - 1. Point-to-point lines between exactly two routers.**
 - 2. Multiaccess networks with broadcasting**
 - most LANs
 - 3. Multiaccess networks without broadcasting**
 - most packet-switched WANs

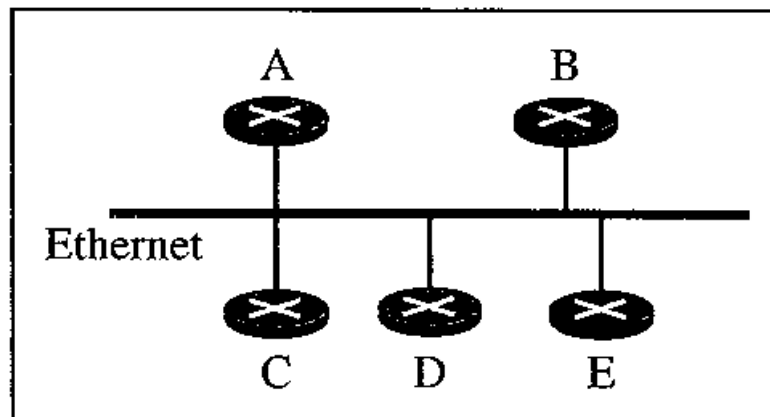
OSPF

Point-to-point lines



A multi-access network is one that can have multiple routers on it.

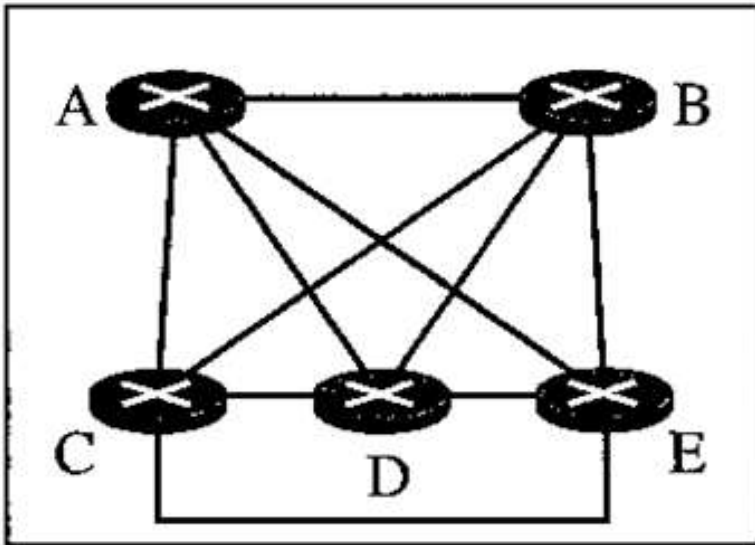
multi-access network



OSPF

Routers on the same LAN are all neighbours

- Therefore we can represent the previous LAN as follows:

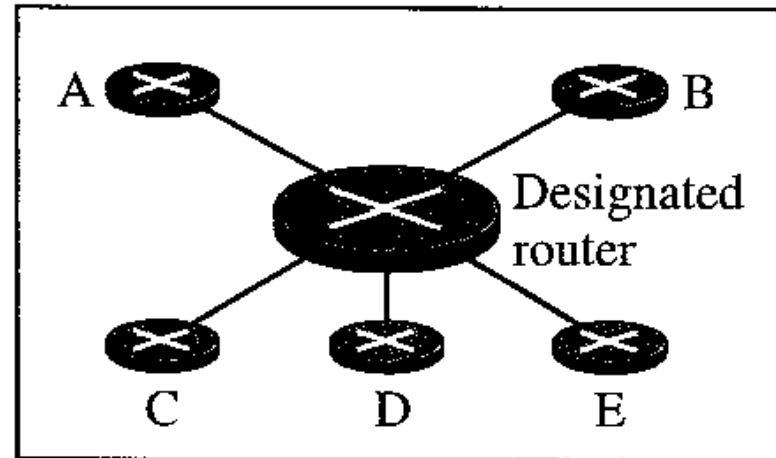
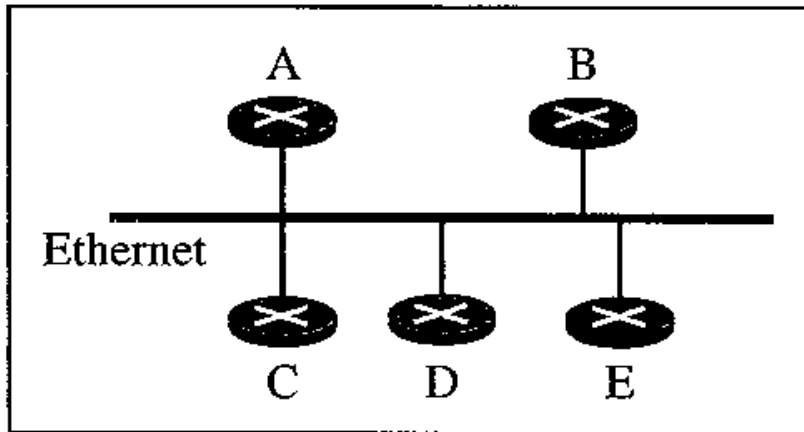


*20 Hello packets will get generated
with this representation*

So it is inefficient to have every router on a LAN talk to every other router on the LAN

OSPF

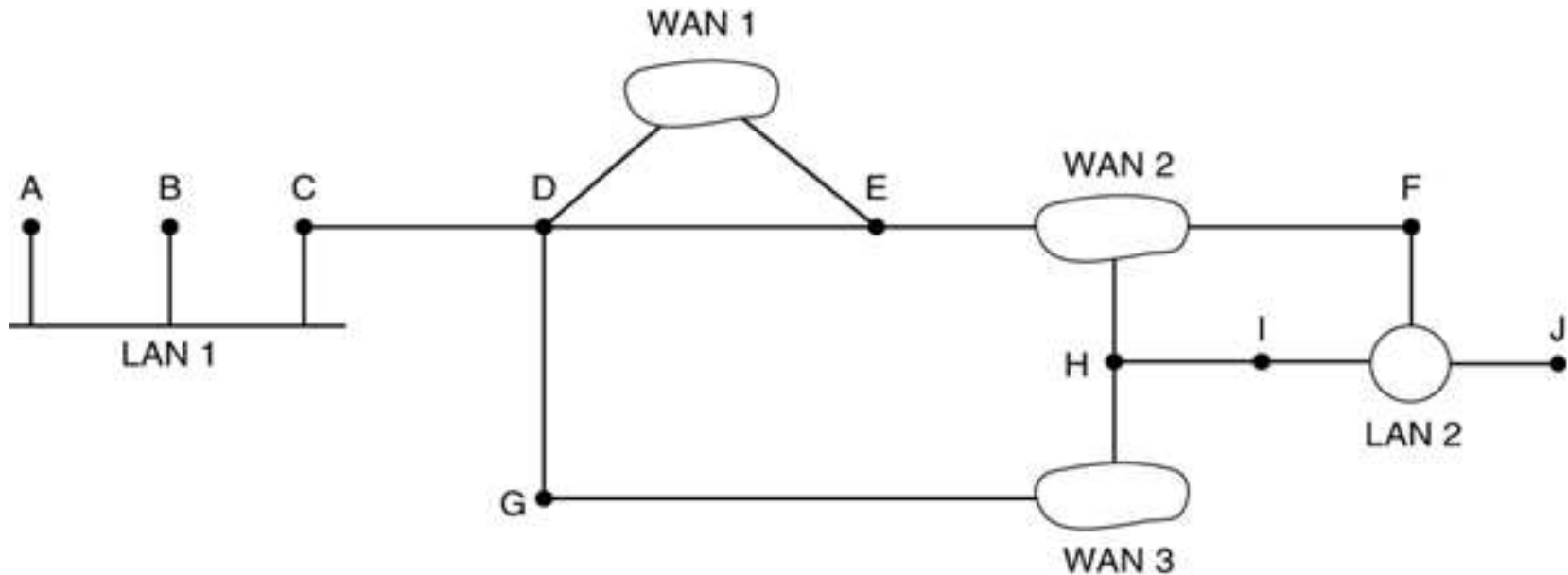
Therefore a better representation is replace the LAN by a node (designated router) for the network itself plus a node for each router.



- Designated router serves as a crossroad between all five routers.
- Only *10 Hello packets will get generated with this representation*

OSPF

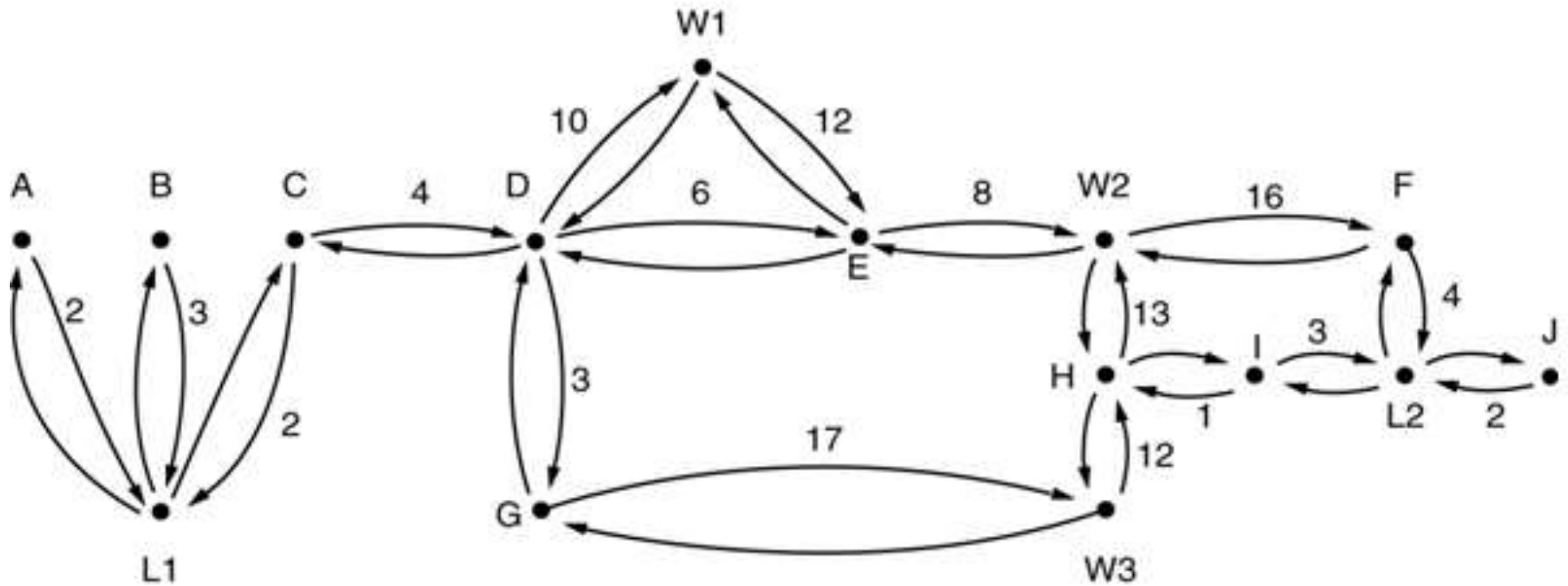
Graphical Representation of an AS – Example



An autonomous system.

OSPF

Graphical Representation of an AS – example



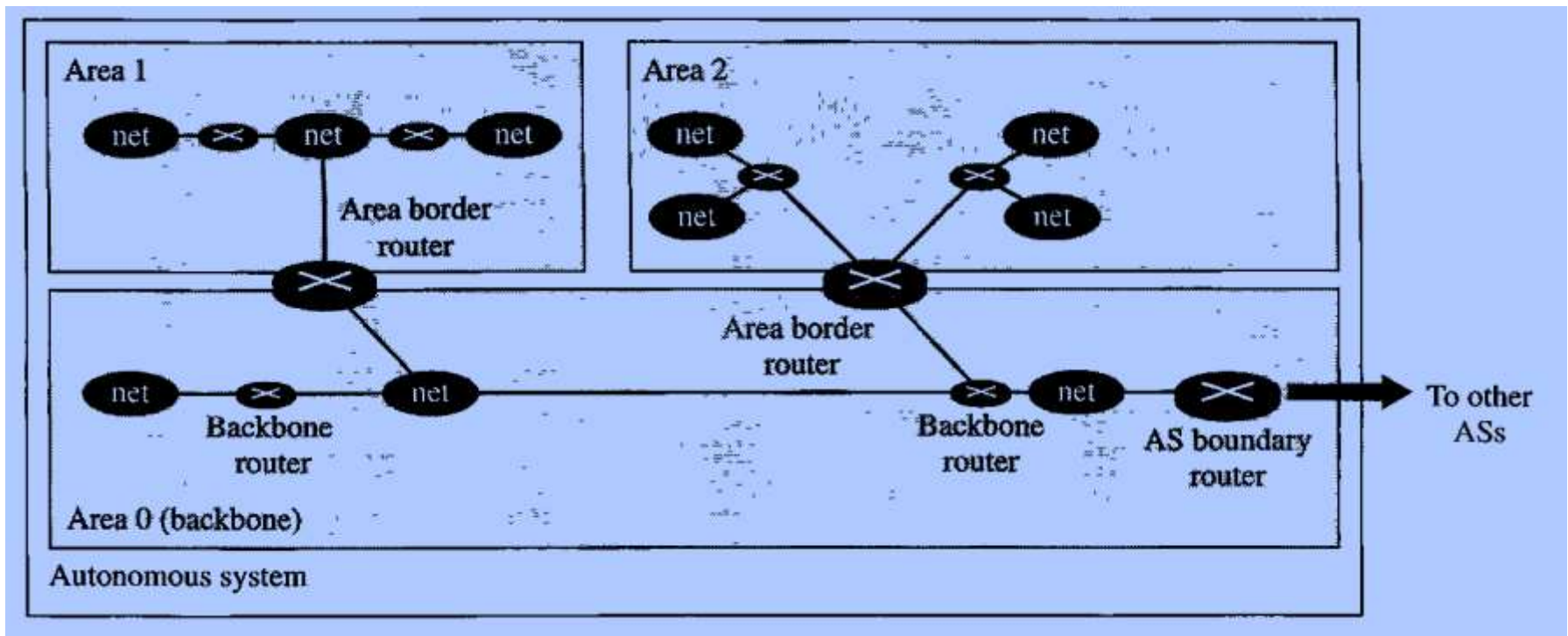
OSPF

Note :

- A serial connection between two routers is represented by a pair of arcs, one in each direction. *Their weights may be different*
- The arcs from the network node to the routers have weight 0 and are omitted from the graph

OSPF

To handle routing efficiently and in a timely manner, OSPF divides an autonomous system into areas



OSPF

- Every AS has a backbone area, called area 0
- All areas are connected to the backbone by tunnels
- A tunnel is represented in the graph as an arc and has a cost
- Each router that is connected to two or more areas is part of the backbone
- Outside an area, its topology and details are not visible

OSPF

- Within an area, each router has the same link state database and runs the same shortest path algorithm.
- The router calculates the shortest path from itself to every other router in the area, including the router that is connected to the backbone.
- A router that connects to two areas needs the databases for both areas and must run the shortest path algorithm for each one separately

OSPF

- During normal operation, three kinds of routes may be needed.

1. intra-area route
2. Interarea route
3. inter-AS route

OSPF

Interarea routing always proceeds in three steps

1. go from the source to the backbone;
2. go across the backbone to the destination area;
3. go to the destination.

Packets are tunneled if required

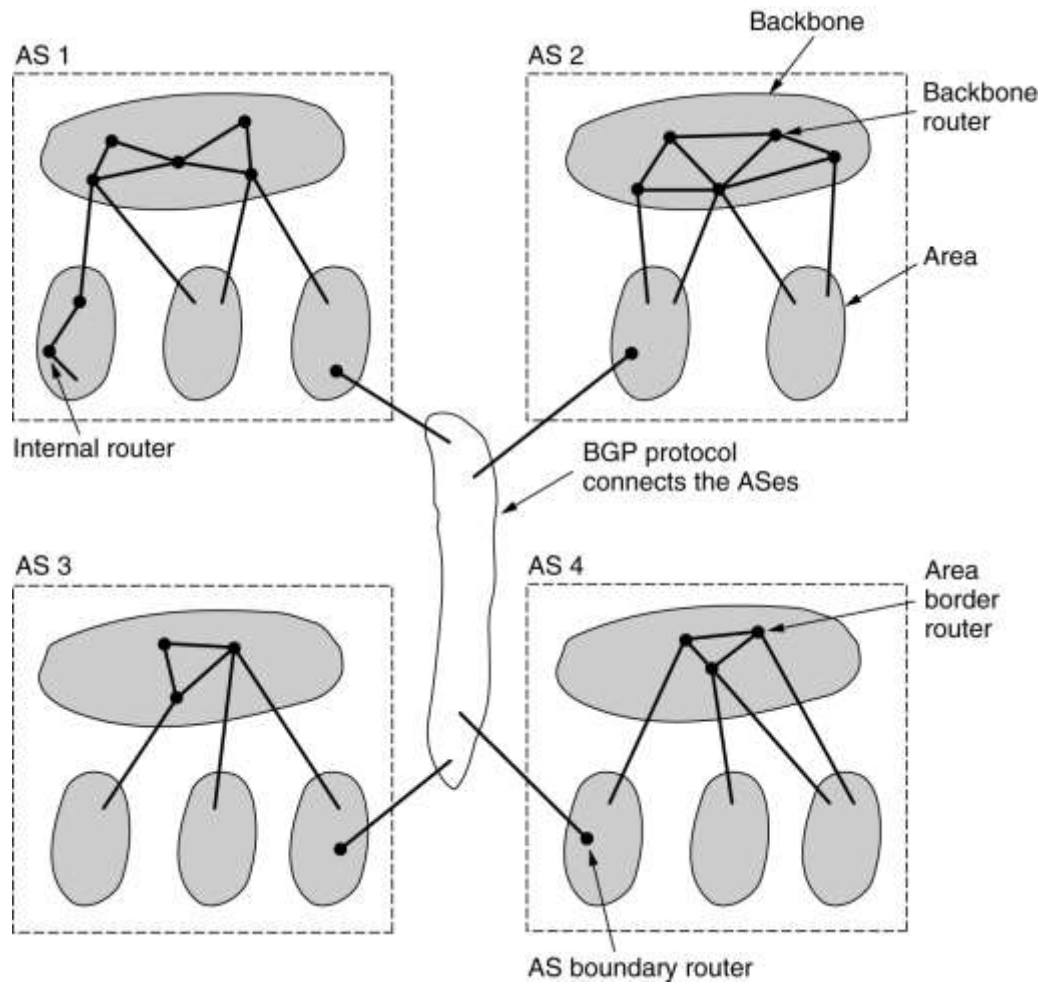
OSPF

OSPF distinguishes four classes of routers:

- 1. Internal routers** are wholly within one area.
- 2. Area border routers** connect two or more areas.
- 3. Backbone routers** are on the backbone.
- 4. AS boundary** routers talk to routers in other ASes.

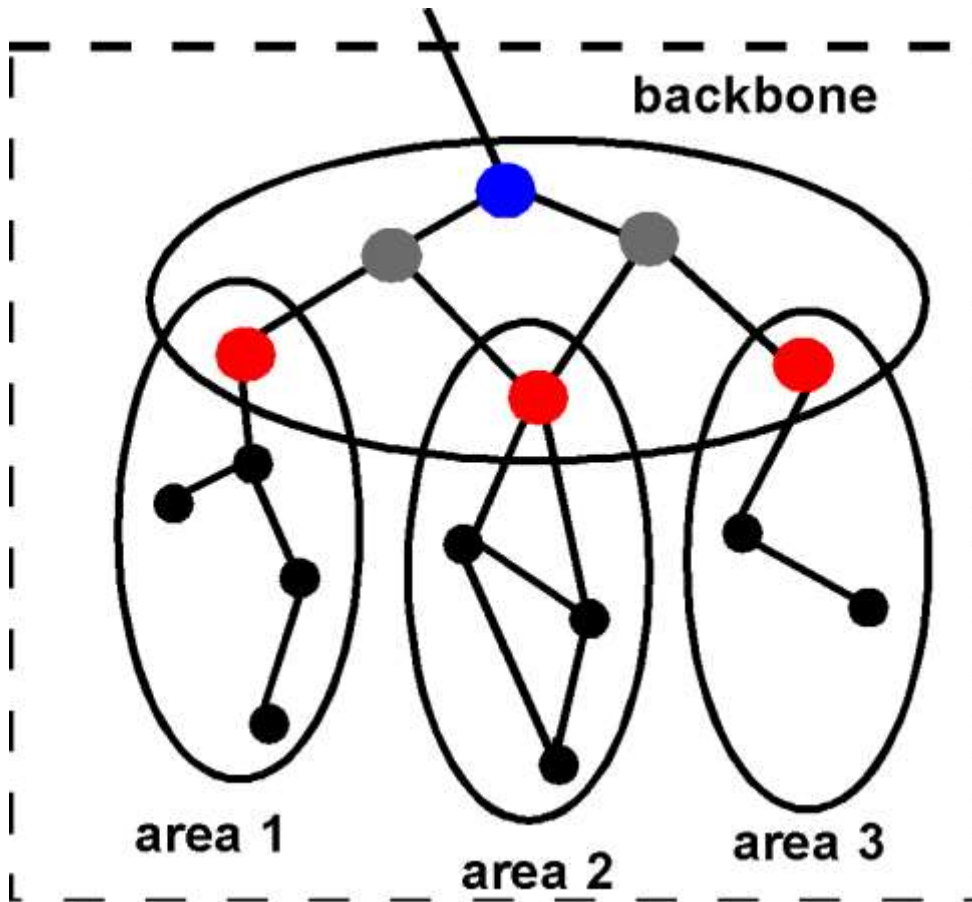
OSPF

four classes of routers:



OSPF

These classes are allowed to overlap



For example : all the border routers are automatically part of the backbone

Routing in OSPF

Learning about neighbours

- Each router sends HELLO messages on all of its point-to-point lines
- multicasts them on LANs
- On WANs, it needs some configuration information to know who to contact.

Routing in OSPF

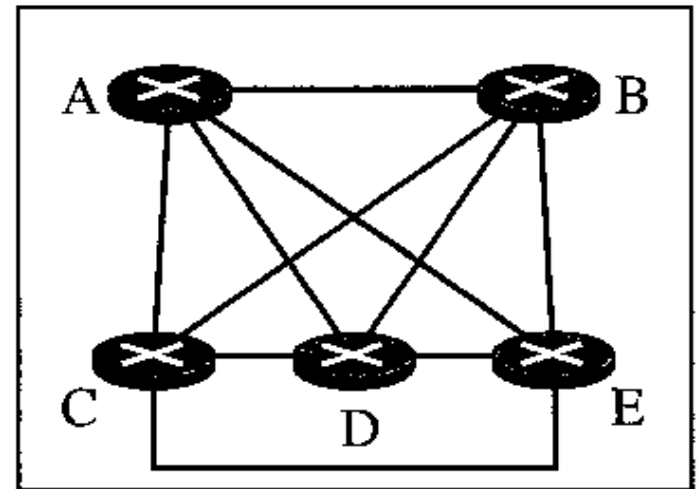
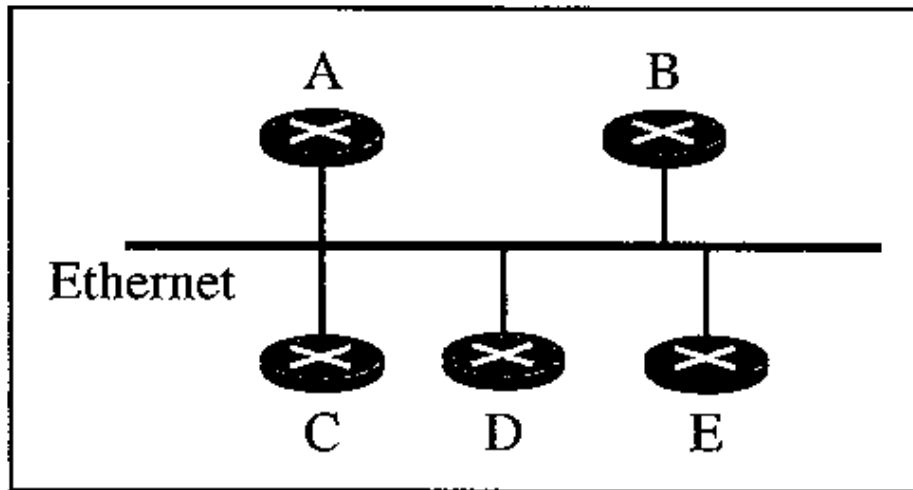
OSPF has five basic steps:

- 1. Learning about neighbours**
- 2. Establishing adjacencies and synchronizing databases**
- 3. Flooding link state information and building routing tables**
- 4. computing shortest paths**

Routing in OSPF

Learning about neighbours

- Routers on the same LAN are all neighbours

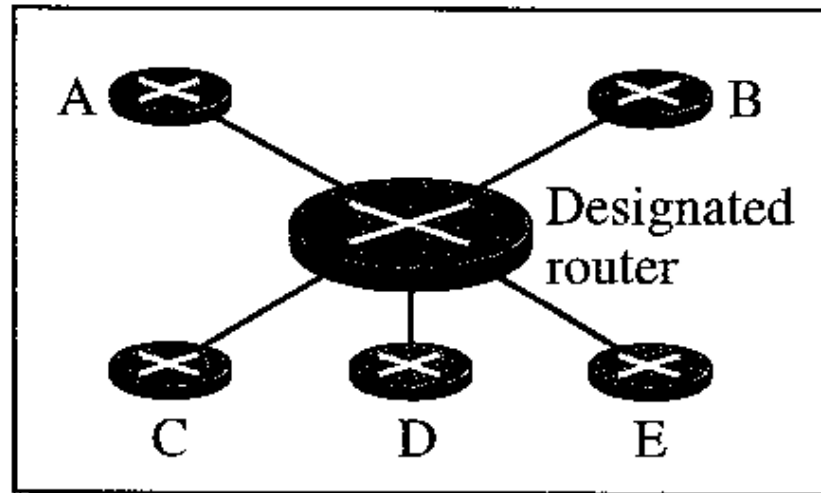


It is inefficient to have every router on a LAN talk to every other router on the LAN.

Routing in OSPF

Learning about neighbours

- To avoid this situation, one router is elected as the designated router



. The designated router is said to be adjacent to all the other routers on its LAN, and exchanges information with them

Routing in OSPF

Establishing adjacencies and synchronizing databases

- OSPF involves establishing ``adjacencies" between a subset of the routers in AS.
- Once two neighbouring routers establish adjacency, they exchange **DATABASE DESCRIPTION** packets to synchronize their link-state databases.

Routing in OSPF

Establishing adjacencies and synchronizing databases

- DATABASE DESCRIPTION messages give the sequence numbers of all the link state entries currently held by the sender
- These sequence numbers help receiver to synchronize with sender's link state database

Routing in OSPF

Flooding LINK STATE UPDATE messages

- Each router periodically floods **LINK STATE UPDATE** messages to each of its adjacent routers.
- A router can request link state information using **LINK STATE REQUEST** messages
- The flooding messages are acknowledged.
- Each message has a sequence number.
 - So a router can see whether an incoming **LINK STATE UPDATE** is older or newer than what it currently has

Routing in OSPF

The five types of OSPF messages

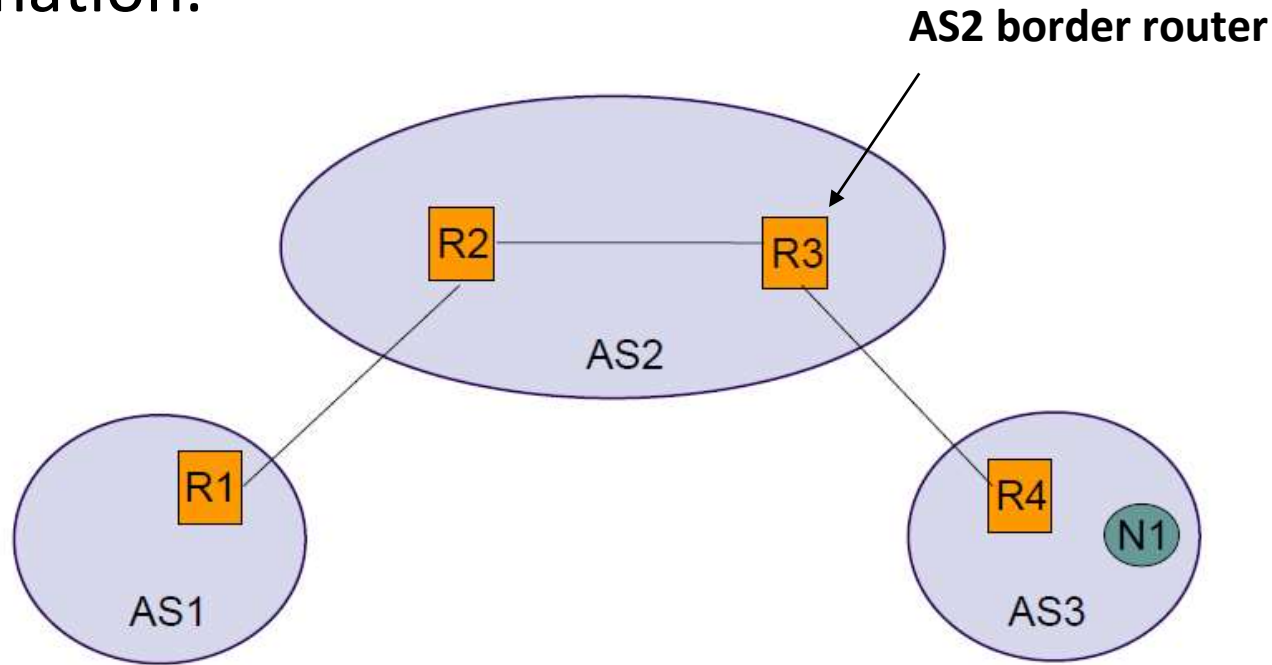
Message type	Description
Hello	Used to discover who the neighbors are
Link state update	Provides the sender's costs to its neighbors
Link state ack	Acknowledges link state update
Database description	Announces which updates the sender has
Link state request	Requests information from the partner

BGP- Border Gateway Protocol

The Exterior Gateway Routing Protocol

BGP

- The BGP is an interAS (inter-domain) routing protocol to enable two different ASs to exchange routing information.



BGP

- Because each AS has its own administrative control, the focus of EGPs is **more on policy issues than on path optimality.**
- **Typical policies involve political, security, or economic considerations.**

For example :

Do not use the United States to get from British Columbia to Ontario.

BGP

Types of Autonomous Systems

: stub, multiconnected, and transit

1. Stub AS. ((either a source / sink)

- has only one connection to another AS
- can send / receive traffic to/from other ASes
- Data traffic, however, cannot pass through a stub AS

e.g : small corporation

BGP - *Types of Autonomous Systems*

2. multiconnected ASs

- has more than one connection to other ASes,
- It can receive /send data traffic from/to more than one AS
- It does not allow data coming from one AS and going to another AS to pass through.

e.g. : a large corporation

BGP - *Types of Autonomous Systems*

3. Transit AS (such as backbones)

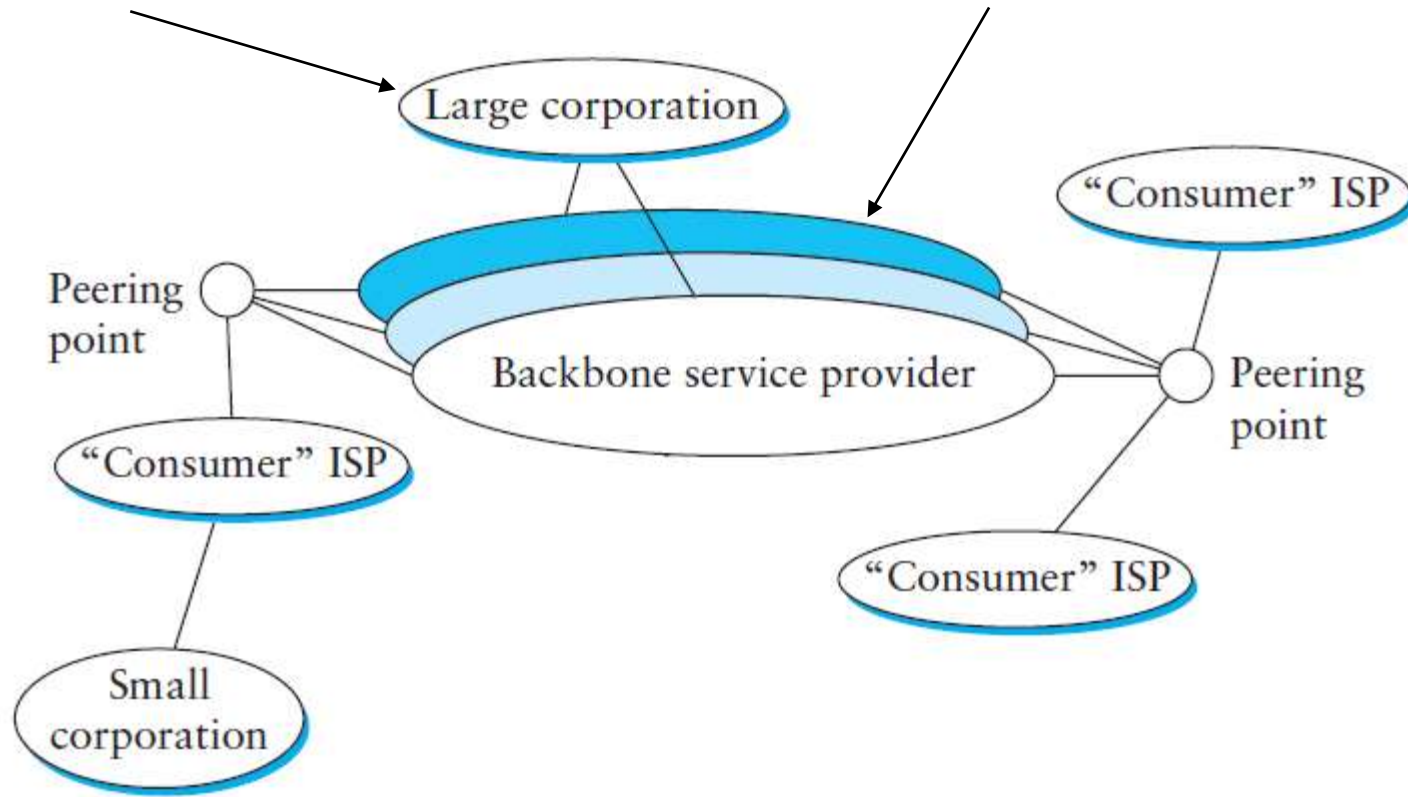
- These are willing to handle third-party packets,
- possibly with some restrictions
- and usually for pay.

e.g. : national and international ISPs (Internet backbones).

BGP

multiconnected AS

Transit AS



Stub AS

Types of Autonomous Systems

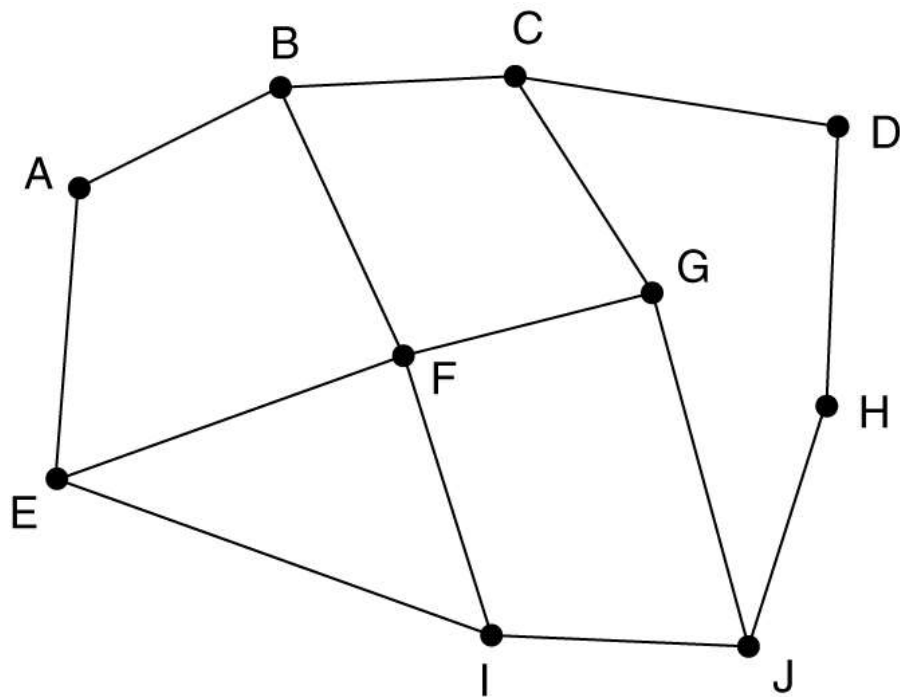
BGP routing

- **BGP is a distance vector protocol**, but quite different from most others such as RIP.
 - Each BGP router keeps track of the path used
 - Each BGP router tells its neighbours the exact path it is using.

BGP routing

Example

Suppose that F wants the best path to get to D. When the neighbours give it routing information, they provide their complete paths as shown:



(a)

Information F receives
from its neighbors about D

From B: "I use BCD"

From G: "I use GCD"

From I: "I use IFGCD"

From E: "I use EFGCD"

(b)

BGP routing

Example

- Now router F quickly discards the paths from I and E, since these paths pass through F itself.
- Every BGP router contains a module that scores routes to the given destination.
- Any route violating a policy constraint automatically gets a score of infinity
- The router then adopts the route with the shortest distance

BGP routing

BGP easily solves the count-to-infinity problem. How?

- Because F quickly discards the paths from I and E (these paths pass through F itself) , **I and E cannot fool F**